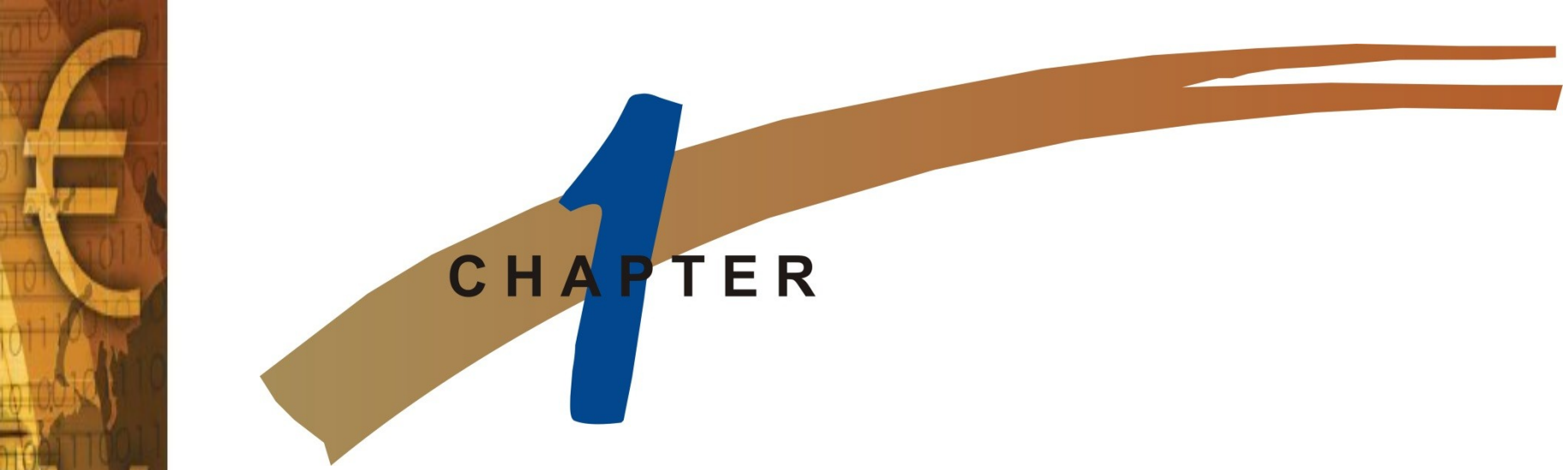


The Dawn of a Maturing Industry

ELECTRONIC COMMERCE
From Vision to Fulfillment

Third Edition

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The focus of this chapter is on several learning objectives

- Conceptual understanding of e-commerce, e-business and e-strategy
- Advantages and Issues in E-Commerce
- Value-chain and supply-chain management and how they relate to e-commerce and e-business
- Business models of the e-environment
- A trend toward integrating e-commerce

What Is E-commerce?

- **Communications Perspective :** the ability to deliver products, services, information, or payments via networks.

- **Interface View :** E-commerce means information and transaction exchanges between:
  - Business-to-Business (B2B)
  - Business-to-Consumer (B2C)
  - Consumer-to-Consumer (C2C)
  - Business-to-Government (B2G)

What Is E-commerce?

- **Business Process :** E-commerce means activities that support commerce electronically by networked connections.
- **Online Perspective :** E-commerce is an electronic environment that allows sellers to buy and sell products, services, and information on the Internet.
- **A Structure :** E-commerce deals with various media: data, text, Web pages, Internet telephony, and Internet desktop video.
- **A Market :** E-commerce is a worldwide networked Market.

E-business

- Sometimes used interchangeably to refer to e-commerce
- E-business covers online transactions, but also extends to all Internet based interactions with business partners, suppliers and customers such as:
  - selling direct to consumers,
  - manufacturers and suppliers;
  - monitoring and exchanging information;
  - auctioning surplus inventory;
  - collaborative product design.
- aimed at improving or transforming business processes and efficiency.

E-commerce Is Not E-business

- *E-business* **- the conduct of business on the Internet, in supply-chain planning, tracking, fulfillment, invoicement, and payment.**
  - **Includes buying and selling as well as servicing customers and collaborating with business partners**
  - **Electronic information is used to boost performance and create value by forming new relationships between and among businesses and customers.**
  - **One example of e-business is SAP (www.sap.com)**

E-commerce Versus E-business

- **E-commerce**
  - Selling goods and services on the retail level with anyone, anywhere, via the Internet
  - Greater efficiency and effective exchange of goods and services
  - Exchange based upon *transaction*
    - A block of information exchange between the merchant and its customers via the corporate Web site
    - **www.amazon.com**

- **E-business**
  - Connecting critical business systems and constituencies directly via the Internet

The Drivers for E-commerce

- **Digital Convergence -** digital devices communicating with one another
- **Anytime, anywhere, anyone -** e-commerce is available to anyone (24/7)
- **Changes in Organizations -** today's businesses empower frontline workers to do the kind of work once performed by junior management
- **Increasing pressure on operating costs and profit margins -** global competition and the proliferation of products and services worldwide have added pressure on operating costs and profit margins
- **Demand for customized products and services -** mass customization puts pressure on firms to handle customized requests on a mass-market scale

Myths about E-commerce

- Setting up a Web site is easy.
- E-commerce means no more mass marketing.
- E-commerce means a new economy.
- E-commerce is revolutionary.
- E-commerce is a commercial fad that crashed in 2000.
- All products can be sold online using identical business models.
- Build it and they will come.
- The middleman is out.

Advantages through E-commerce

- **Lower cost to the E-merchant**
  - Cost effective
  - Reduces logistics problem
  - Give small company equal exposer as the big companies
  - Lower transaction processing charge
- **Economy**
  - No rental of physical storage space, insurance or infrastructural investment
  - Only need a good idea, unique product and well designed storefront
- **Higher margins**
  - Businesses gain more control and flexibility when manual transaction is done electronically
  - Low transaction processing charge

Advantages through E-commerce

- **Better customer service**
  - Web based customer service make user happy
  - Let user serve him/herself save time and money
- **Quick comparison shopping**
  - E commerce allows comparison shopping using automated shopping assistant program called s*hopbot*
  - shopbots finds best deals from web storefronts
- **Productivity gains**
  - Weaving the web tech. throughout the organization means improved productivity
- **Teamwork**
  - Supports user collaboration in exchanging information and work on solution
  - More interaction mean better overall result.
  - improved communication with its suppliers, vendors and business partners

Advantages through E-commerce

- **Information sharing, convenience, and control**
  - Improves information sharing between merchants and customer and promote quick, just-in-time deliveries
  - Open 24x7, no traffic jam, do not have to carry shopping bags
  - Business can give more control to his customer managing his account
- **Customization**
  - Digital products are highly customizable
  - Easy to recognize, reverse or edit
  - With the information about the customer products can be customized and matched to individual needs

Issues and Constraints of E-commerce

- The cost factor
- Security
- System and data integrity
- System scalability
- E-commerce is not free
- Fulfillment and customer relations problems
- Products people resist buying online
- Cultural, language, and trust issues
- Corporate vulnerability
- Lack of a blueprint for handling E-commerce
- High risk of Internet start-up

Benefits and Limitations of the Internet

- The Internet is the enabler of the e-commerce
- Marketing and selling products and services
- Doing business fast
- Gathering opinions and trying out new ideas
- Leveling the playing field
- Promoting a paper-free environment
- Providing superior customer service and support resources
- Efficiency and unequaled cost-effectiveness
- Supporting managerial functions, spreading ideas, ease of technical support
- Triggering new business
- Providing Web services

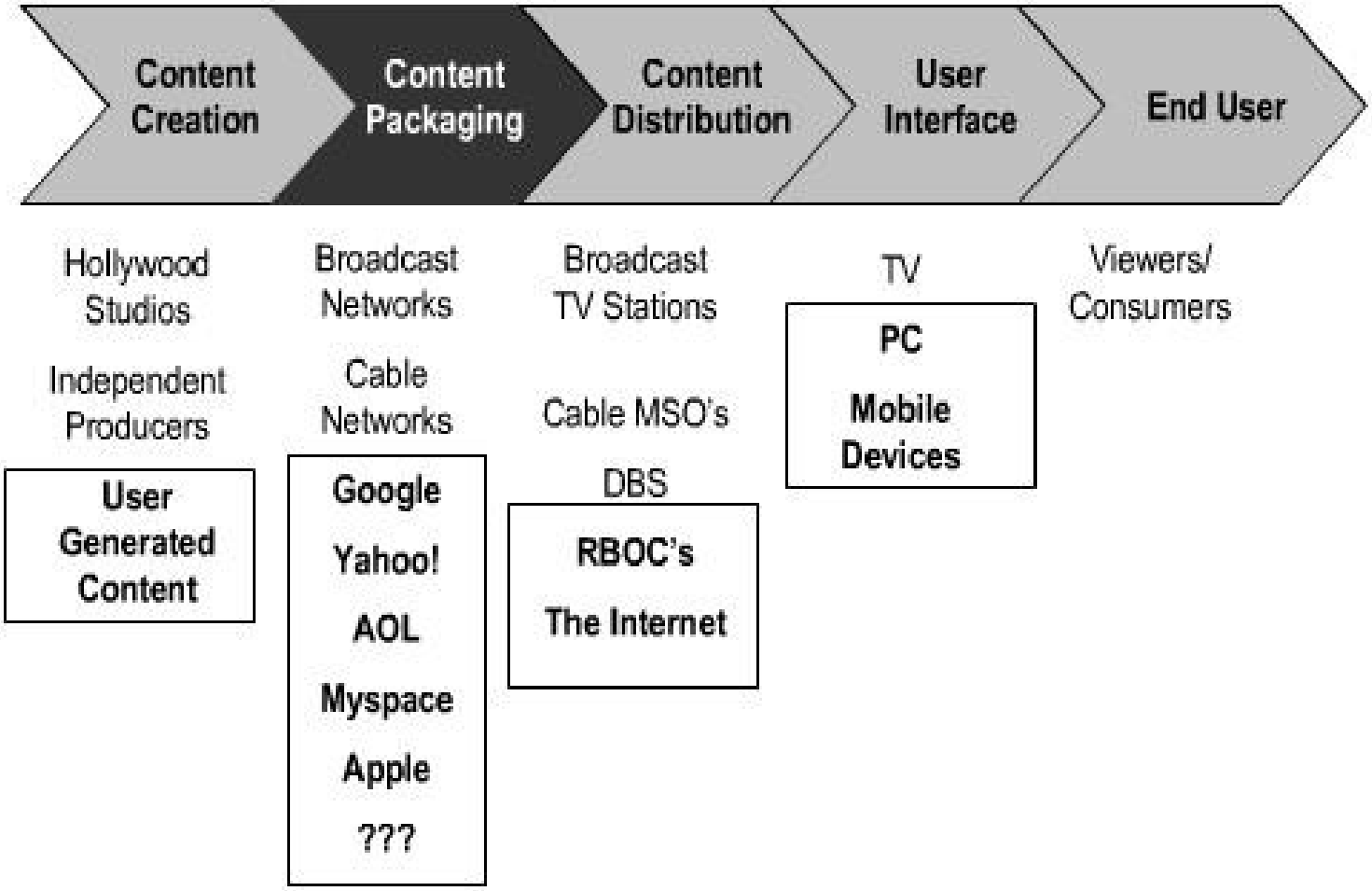
Limitations of the Internet

- Security and privacy
- Fakes and forgeries
- Cyber terrorism
- Problems and stress
- Abuses in the workplace

Value Chain in E-commerce

- **Value Chain:** A way of organizing the activities of a business so that each activity adds value (value-added activity) or productivity to the total operation of the business**.**
- A strategic tool for identifying how the critical components of a business tie together to deliver value for the business across the value-chain process.
- Makes B2C interface operational

Value Chain


Content Creation
Content Packaging
Content Distribution
User Interface
End User
Hollywood Studios
Independent Producers
User Generated Content
Broadcast Networks
Cable Networks
Google
Yahoo!
AOL
Myspace
Apple
???
Broadcast TV Stations
Cable MSO's
DBS
RBOC's
The Internet
TV
PC
Mobile Devices
Viewers/ Consumers

Value Chain Activities



Value Chain Primary Activities

- **Inbound logistics**
  - Procurement Activities: vendor selection, comparative shopping, negotiating supply contract
  - Represents supply side of the business
- **Operations**
  - Conversion of raw material into finished product
  - Machining, packaging, assembly, equipment maintenance, testing etc.
  - Provides added value for the marketing function.
  - Operational activities are the point in the value chain where value is added

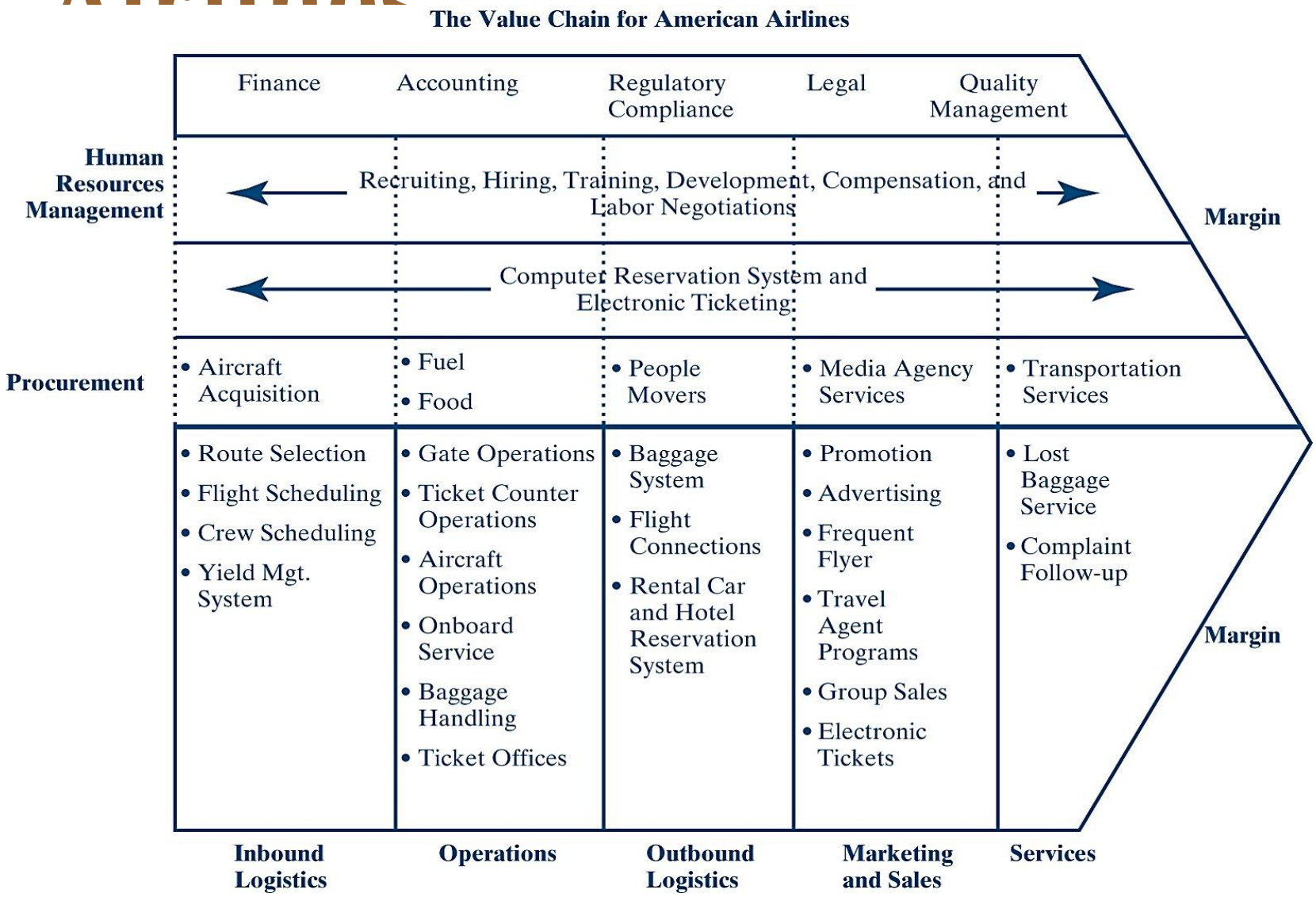
Value Chain Primary Activities

- **Outbound logistics**
  - Storing, distributing and shipping the final product.
  - Involves warehousing, material handling, shipping and timely delivery to the retailer
  - Output of this activity is ties in directly to marketing and sales
- **Marketing and sales**
  - Deals with ultimate customers
  - Includes advertising, product promotion, sales management, identifying customer base and distribution channel
  - Output of this activity triggers
    - increased production,
    - more advertising

Value Chain Primary Activities

- **Service**
  - Focus is on after sales service
  - Includes testing, maintenance, repair, warranty works and replacement parts
  - Output of this activity means
    - satisfied customer,
    - improved image of the product and the business and
    - potential for increased production

Value Chain for American Airlines



Value Chain Support Activities

- **Corporate infrastructure**
  - General management, accounting, finance, planning and legal service, quality management
- **Human resources**
  - Activity to matching right people to the job
  - Involves recruitment, retention, career path development, compensation, training and development etc.
- **Technology development**
  - Improves the products and business processes
  - Output: product quality, integrity and reliability
- **Procurement**
  - Purchasing, ensuring availability of quality raw materials

Analyzing Value Chain Activities

- What type of activity is being performed? Does it add value? Does it ensure the quality of other activities?
- How does the activity add value to the customer?
- Could the same activity be reconfigured or performed in a different way?
- What inputs are used? Is the expected output being produced?
- Is the activity vital? Could it be outsourced, deleted completely, or combined with another activity?
- How does information flow into and out of the activity?
- Is the activity a source of competitive advantage?
- Does the activity fit the overall goals of the organization?

E-commerce Value Chain

- The E-commerce Value Chain means identifying:
  - The competitive forces within the company's e-commerce environment
  - The business model it will use
  - Identifying the value activities that help the e-commerce value chain do its homework
- E-commerce views information technology as part of a company's value chain

Roles for E-commerce

- Reducing costs
- Improving product quality and integrity
- Promoting a loyal customer base
- Creating a quick and efficient way of selling products and services
- Incorporate information technology and telecommunications to improve overall productivity
- Web sites are used to provide and collect information between the e-merchant and the customer
  - Competitive pricing information
  - Invoicing facilitates online payment flows
  - Customized products or orders can be shipped and delivered by independent shippers directly to the customer

E-commerce Application

- Business-to-Consumer (Internet)
- Business-to-Business (Internet and Extranet)
  - Supply-Chain Management
- Business-within-Business (Intranet)
- Business-to-Government (B2G)

Business-to-Consumer (Internet)

- The consumer's use of a merchant's Web storefront or Web site
- Modeled on the traditional shopping experience
- Shopping cart is used to hold goods until the customer is ready to check out
  - Online order form supported by the appropriate software
- **Checkout** is order and payment processing

Business-to-Business (Internet and Extranet)

- **Business-to-Business** e-commerce is industrial marketing. among the processes it handles are fulfillment and procurement
- Companies can conveniently and quickly check their suppliers' inventories or make instant purchases
- B2B often use an extranet: a shared intranet vendors, contractors, suppliers, and key customers

Supply-Chain Management

- Integrating the networking and communication infrastructure between businesses and suppliers
- Having the right product in the right place, at the right time, at the right price, and in the right condition
- Delivery of customer and economic value through integrated management of the flow of physical goods and related information
- Designed to improve organizational processes by optimizing the flow of goods, information, and services between buyers and suppliers in the value chain

Supply-Chain Management – cont'd

- **SCM is:**
  - ***Collaboration*** **among business partners**
  - ***Coordination*** **of logistics for timely delivery of goods or products**
  - ***Cooperation*** **among businesses and suppliers to make sure orders and inquiries are filled correctly**
  - ***Connectivity*** **through networking infrastructure to ensure speed and good response time at all times**

Business-within-Business (Intranet)

- Strictly a “within company” type of information exchange
  - Restricted to internal employees and customers
  - Firewalls to keep out non employees
  - E-mail replaces paper for the communication of messages, order acknowledgement and approvals, and other forms of correspondence
  - The intranet becomes a facilitator for the exchange of information and services among the departments or divisions of a company
  - Different departments with different PCs or local area networks can interact on an intranet

Benefits of an Intranet

- Low development and maintenance costs
- Environmentally friendly because it is company-specific
- Availability and sharing of information
- Timely, current information
- Quick and easy dissemination of information

Business-to-Government (B2G)

- Online procurement
- Ensuring transparency, reducing cost
- The government market is similar to B2B.

- difficulties
  - Changing the status quo in government is not so easy
  - Changes to employee tasks and job restructuring often create resistance
  - Tax savings potential is not easily recognized
  - Committing to technology means constant need for upgrades and additional costs

Chapter Summary

- Electronic commerce (EC) is the ability to deliver products, services, information, or payments via networks such as the Internet and the World Wide Web.
- Electronic business connects critical business systems directly to key constituents
- The rise of specialized Web sites and Web logs (blogs) generated opportunities to read and write on a vast array of topics
- Several drivers promote EC: digital convergence
- Advantages of EC

Chapter Summary (continued)

- Limitations of EC
- A value chain is a way of organizing the activities of a business so that each activity provides added value or productivity to the total operation of the business.
- The transaction life cycle includes three major e-commerce applications: Business-to-Consumer (B2C), Business-to-Business (B2B), and Business-within-Business.
- An intranet wires the company for information exchange.
- Success in the E-commerce field depends on attracting and keeping qualified technical people and managerial talent.
- There are several types of specialized Web sites on the Internet. Each site is based on a business model as a way of doing business to sustain a business - generated revenue.